

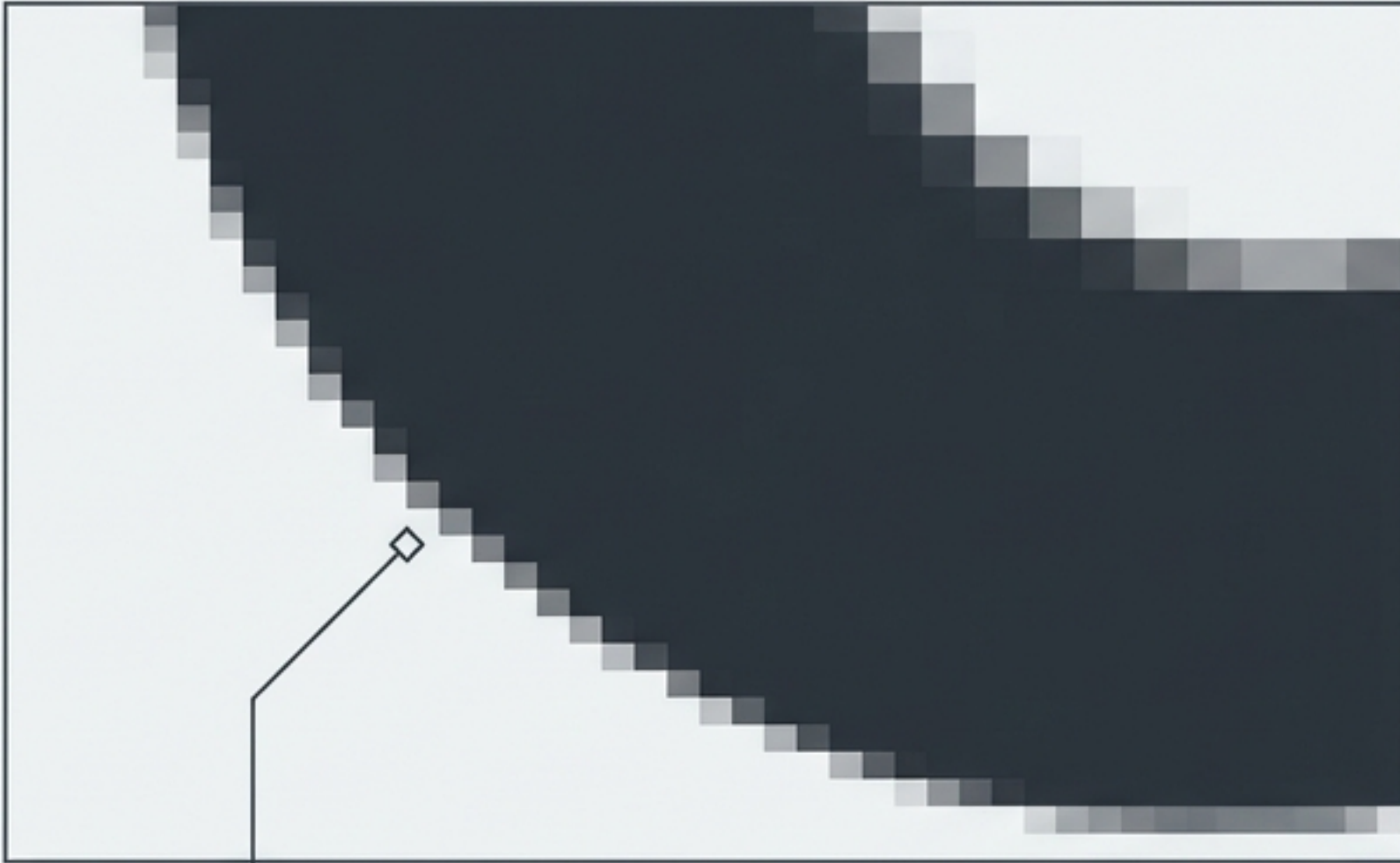


The Inkscape Blueprint

A Masterclass in Vector Design for KS4

Building objects, not painting pixels.

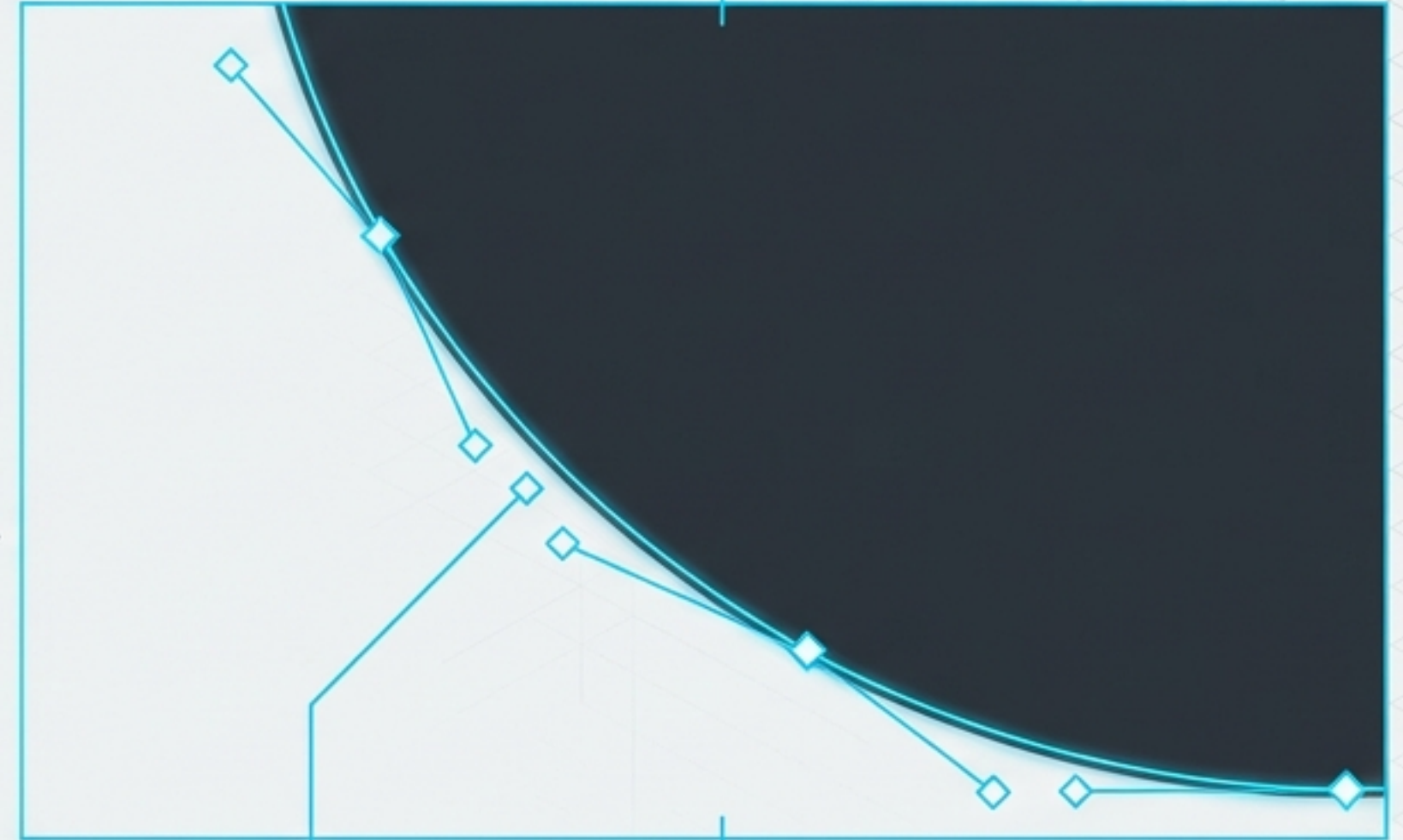
The Pixel World



Raster Graphics

Made of a grid of pixels. Loses quality and goes blurry when resized.

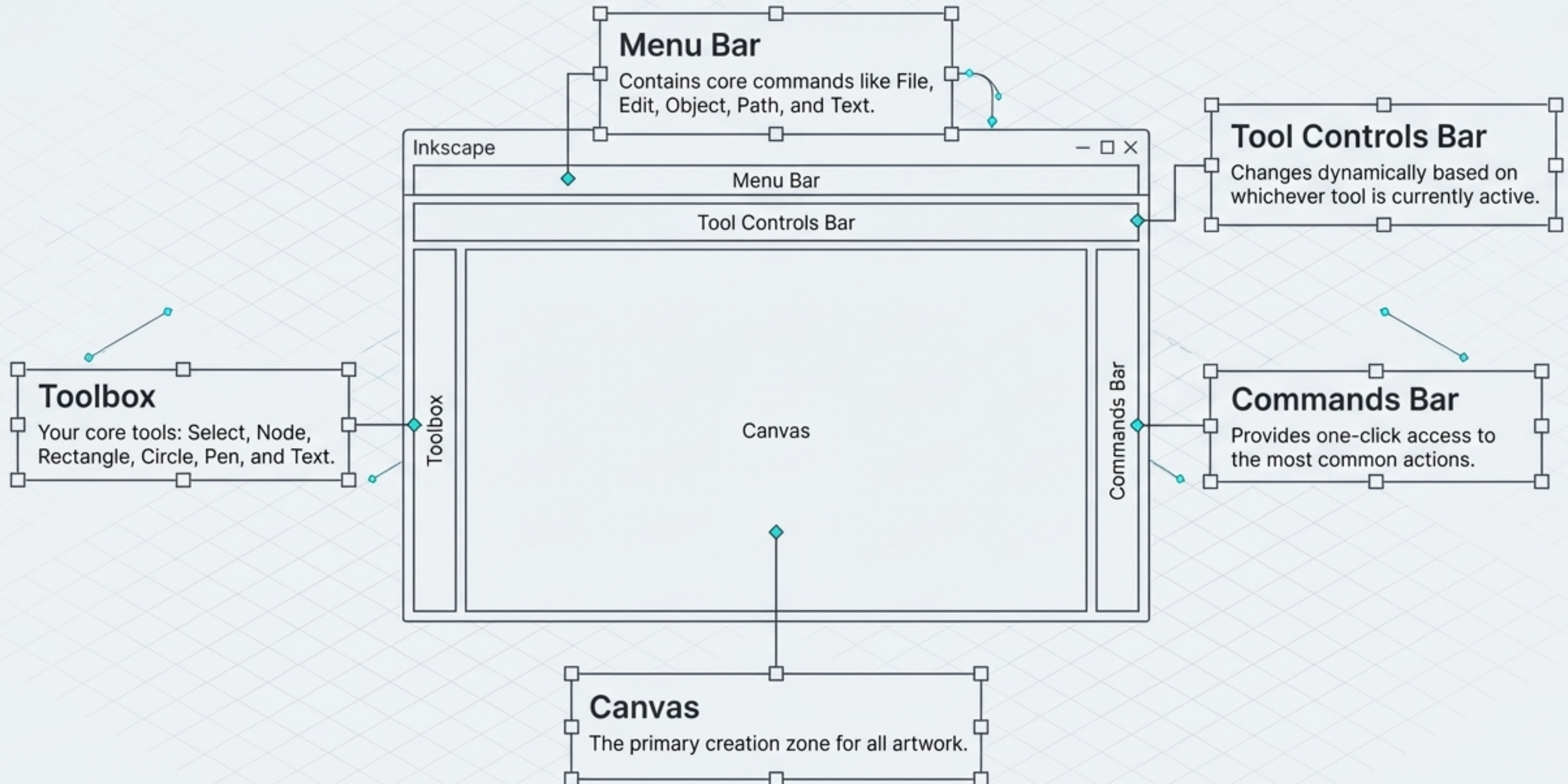
The Vector World



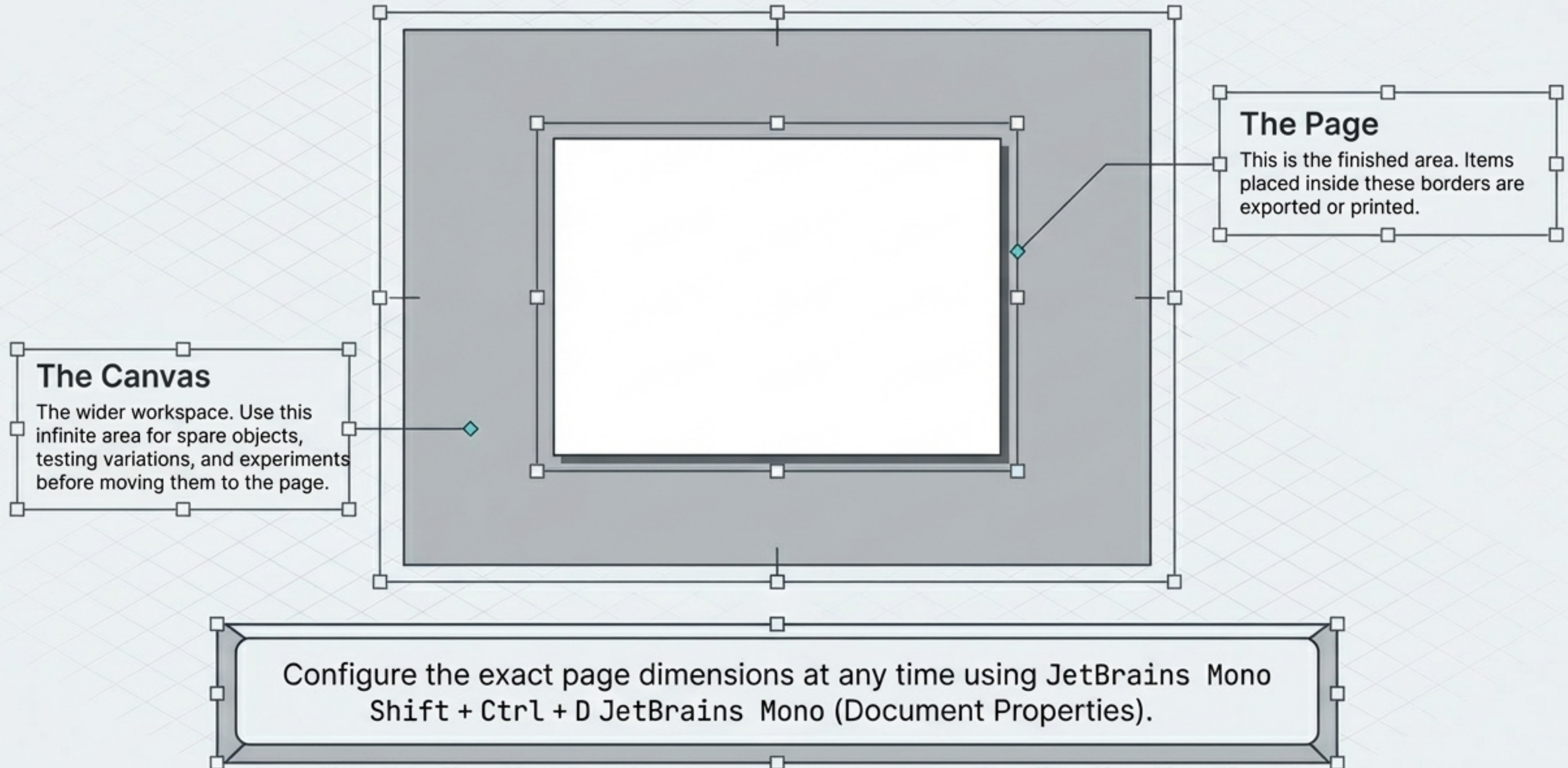
Vector Graphics

Made of lines, shapes, and paths. Infinitely scalable without losing quality. Essential for logos, icons, posters, packaging, and interface layouts.

Anatomy of the Workspace



The Finished Area vs. The Working Area



The Basic Building Blocks

1. Create



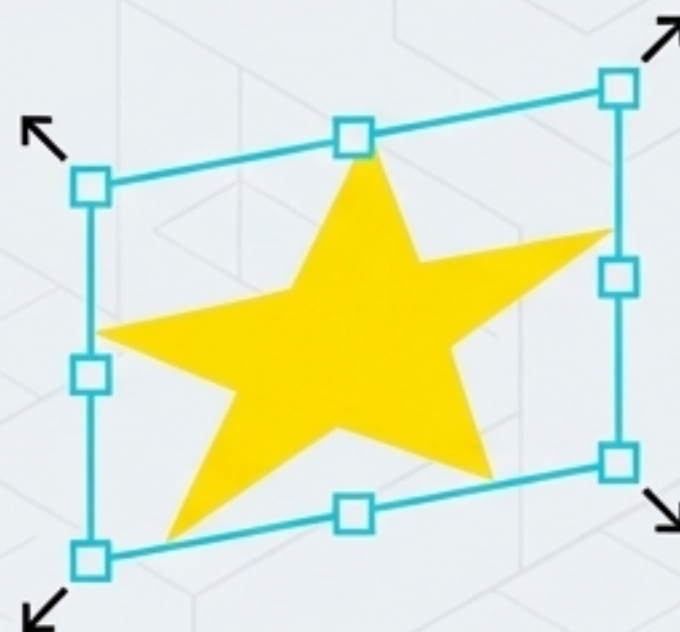
Start with basic shape tools. Building with rigid rectangles, circles, and stars is cleaner and easier than drawing freehand.

2. Select



The Select tool is the basis of everything. Click an object to select it, then click and drag to move it anywhere.

3. Transform



Use the corner handles on the selection box to precisely resize and rotate your objects into the perfect position.

Styling the Objects: Fill vs. Stroke



Fill

The solid inside colour of an object.



Stroke

The outline colour and thickness applied to the edge.

Controlling both fill and stroke independently is essential for poster design, logos, and infographics. Quickly copy styles from one object to another using JetBrains Mono `Shift + Ctrl + V` JetBrains Mono (Paste style only).

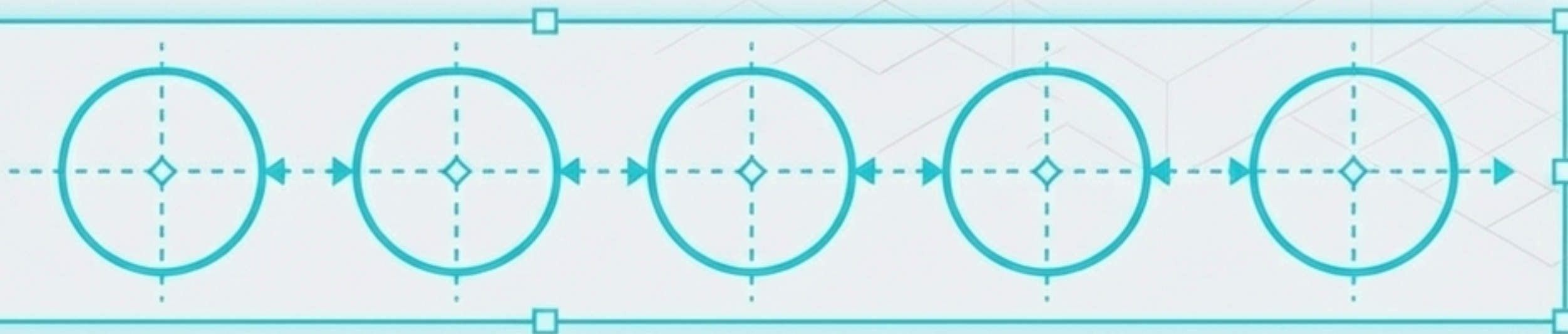
The Rule of Duplication and Alignment

The Amateur Way



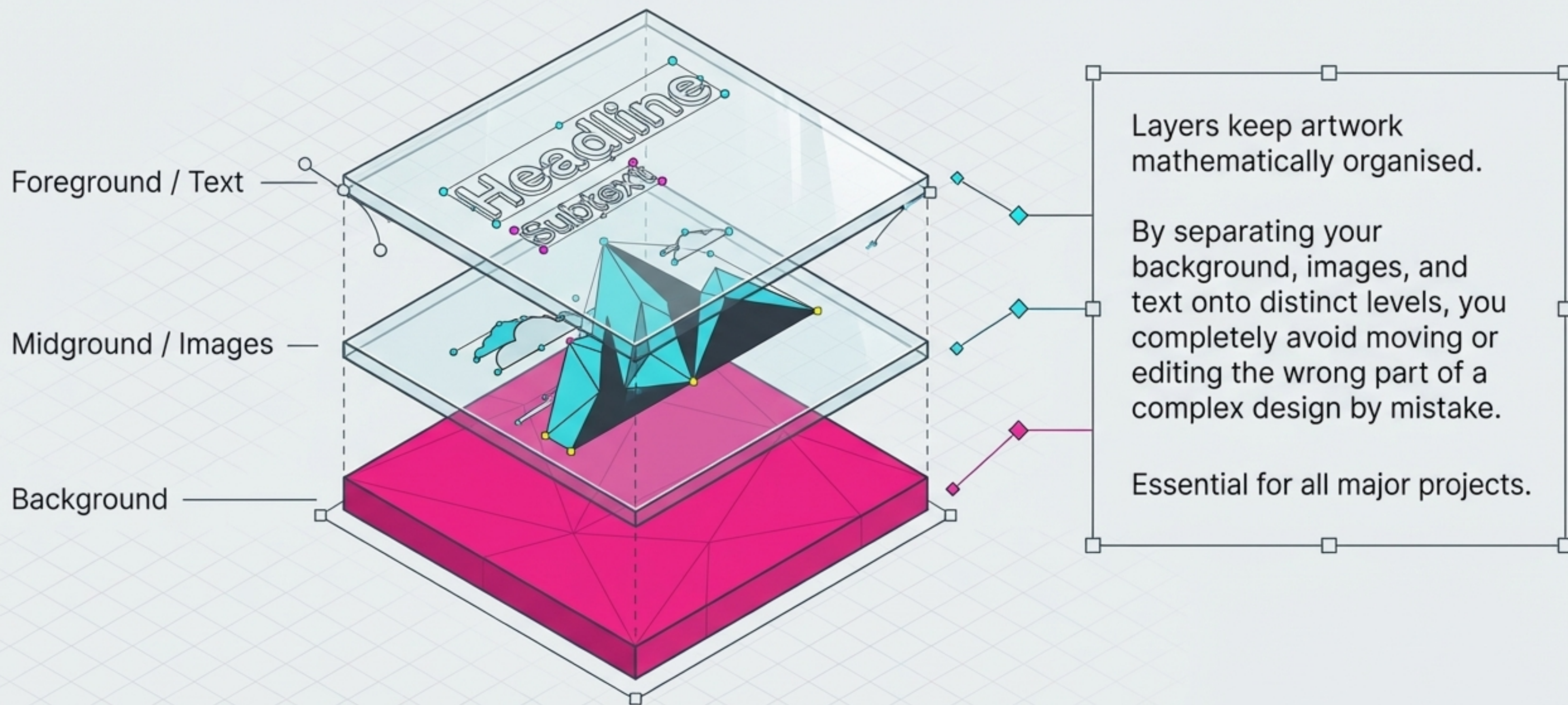
Redraw from scratch → Drag items by eye → Result: Inconsistent, messy, and slow.

The Pro Way

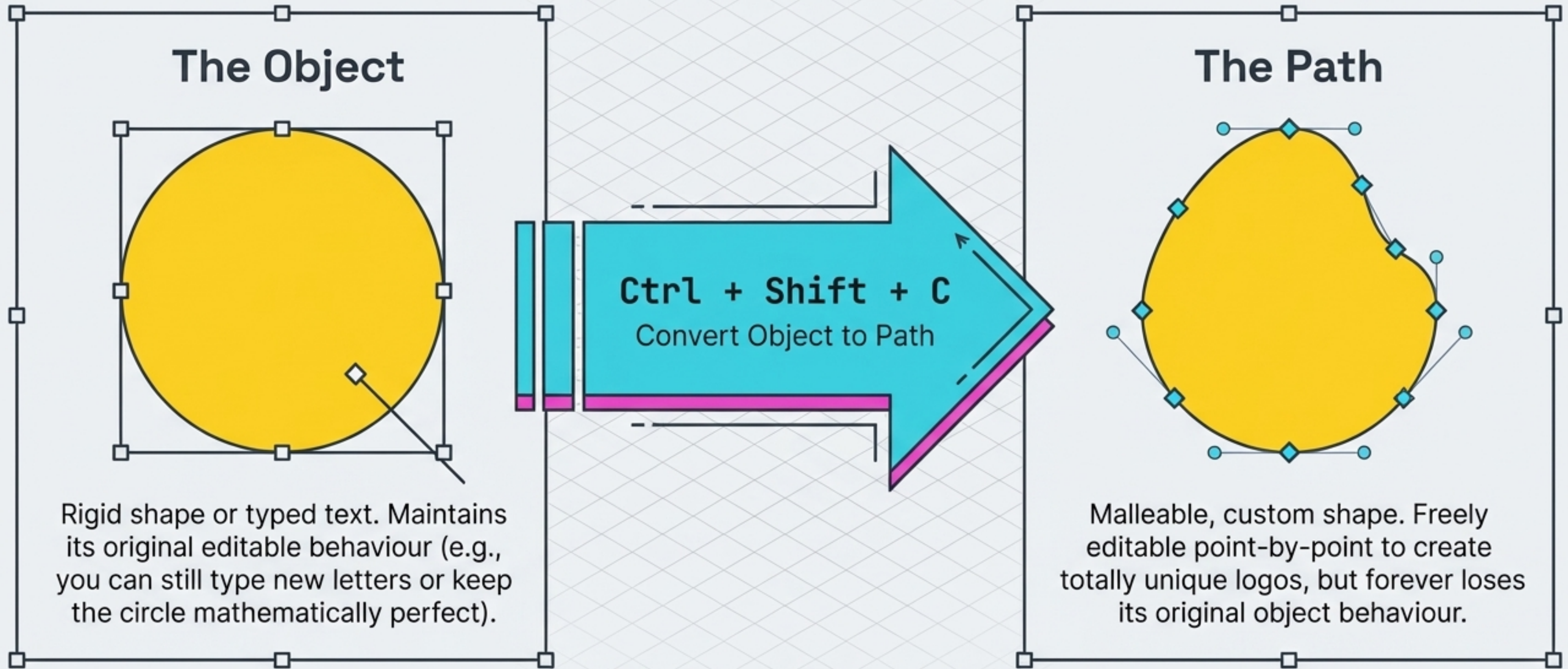


Select object → Press `Ctrl + D` (Duplicate in place) → Press `Shift + Ctrl + A` (Align and Distribute) → Result: Neat, perfectly spaced, professional layouts instantly.

Stacking Order: Working with Layers

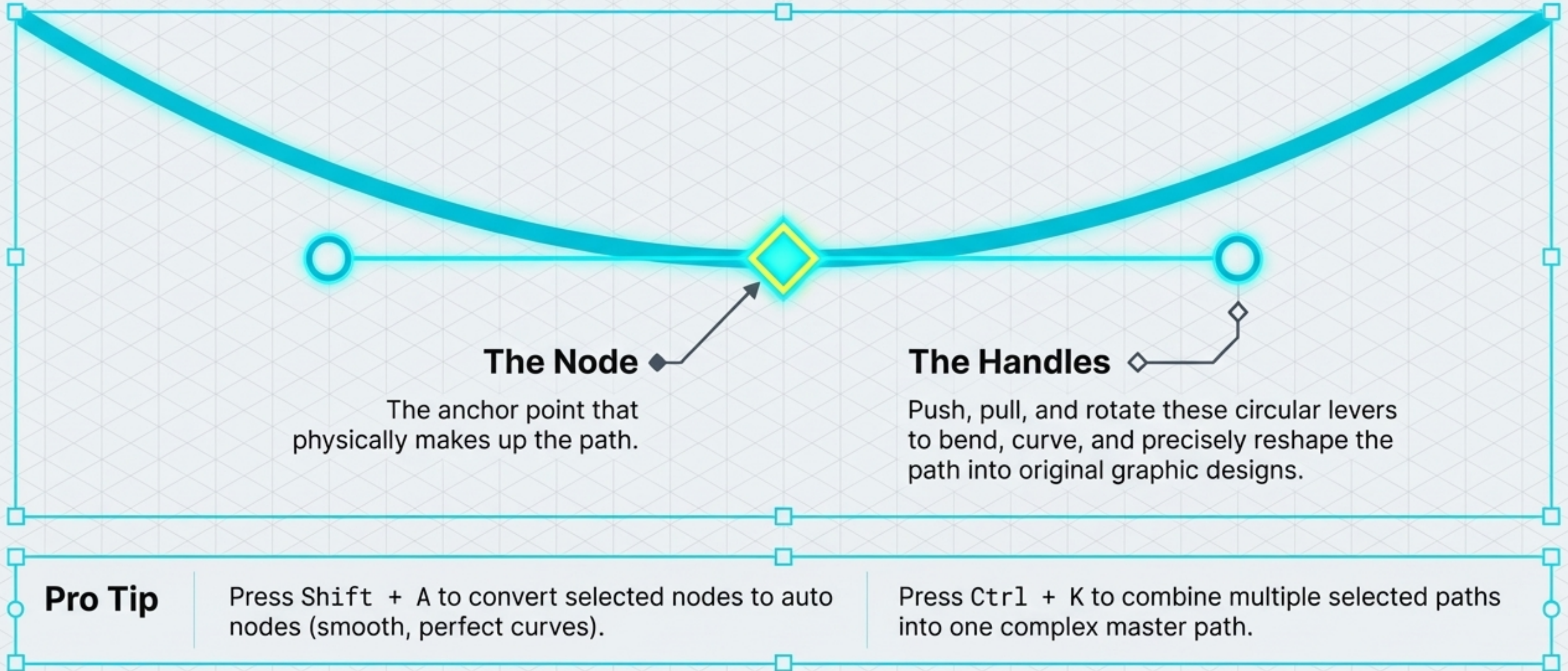
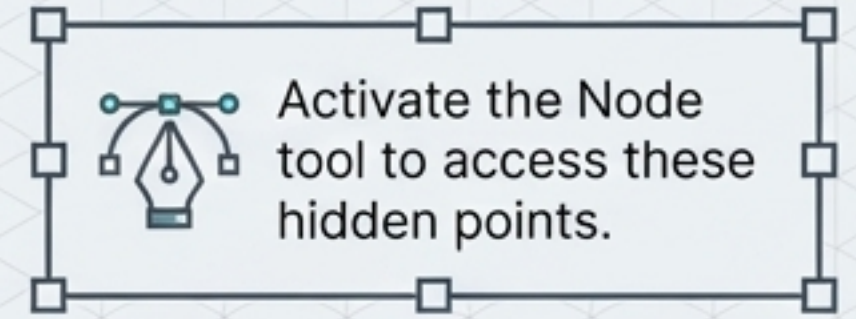


The Conceptual Shift: Object to Path



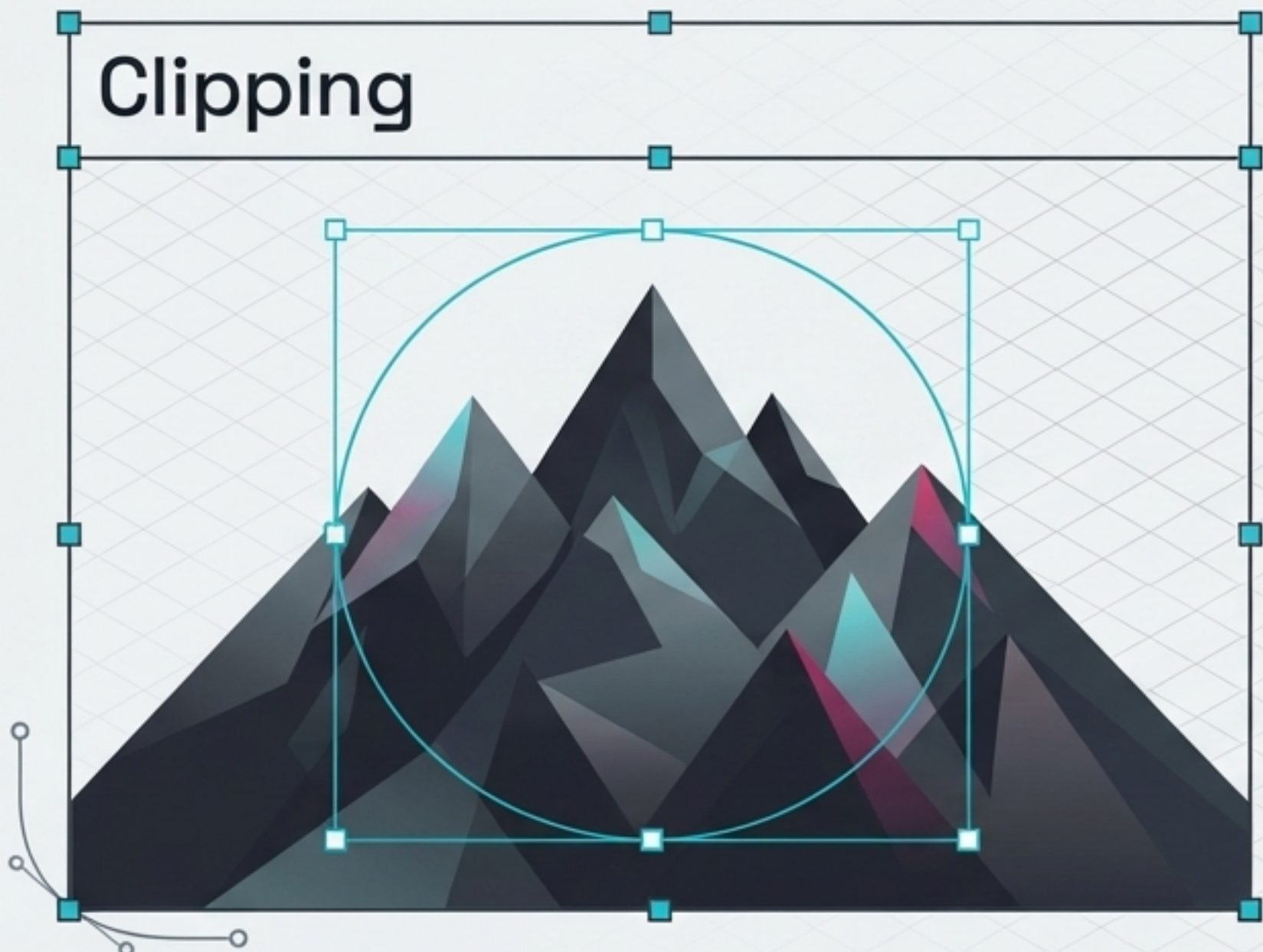
Warning: Don't convert text to paths too early! Save versions carefully.

Precision Control: Editing Nodes



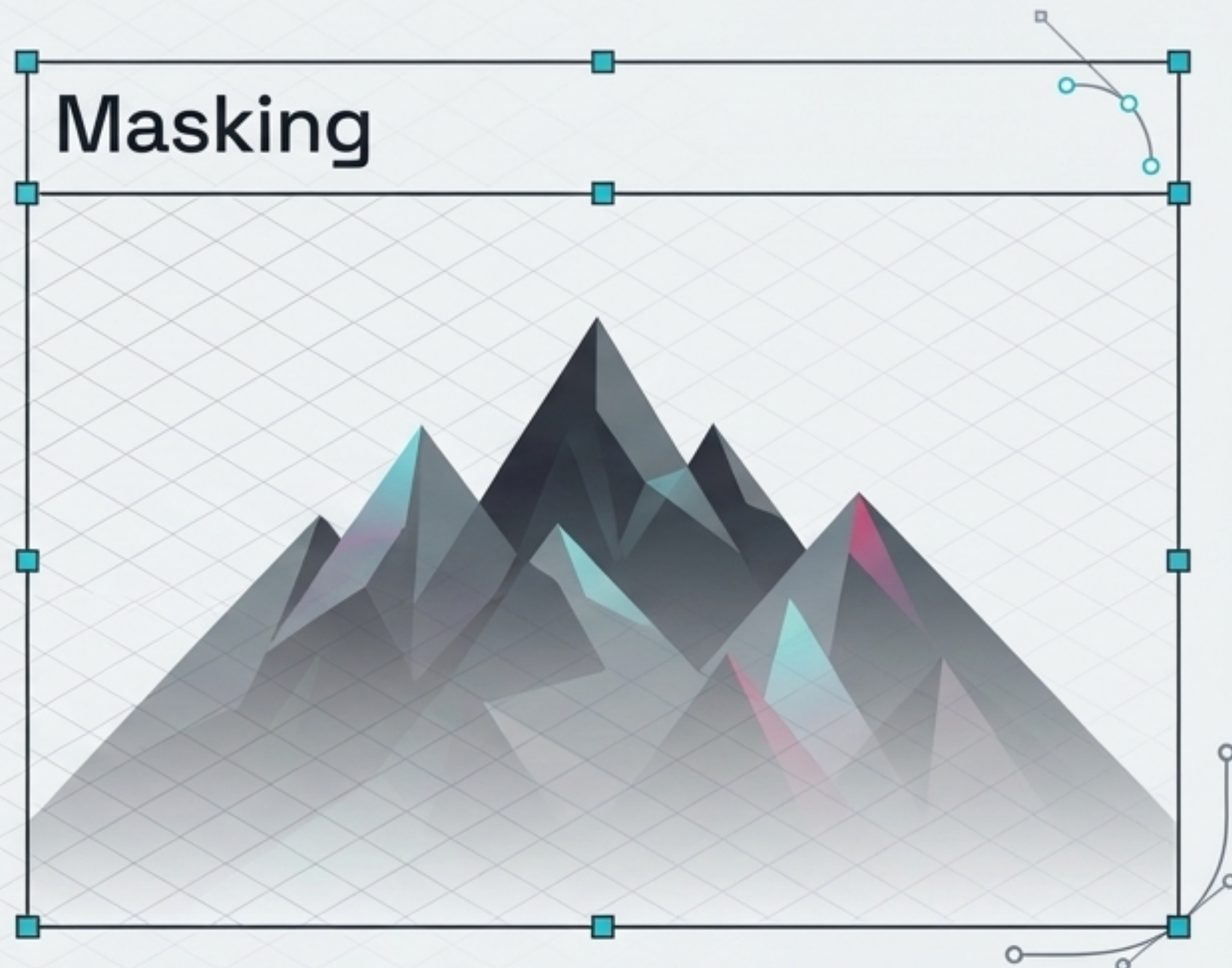
Visual Effects: Clipping vs. Masking

Clipping



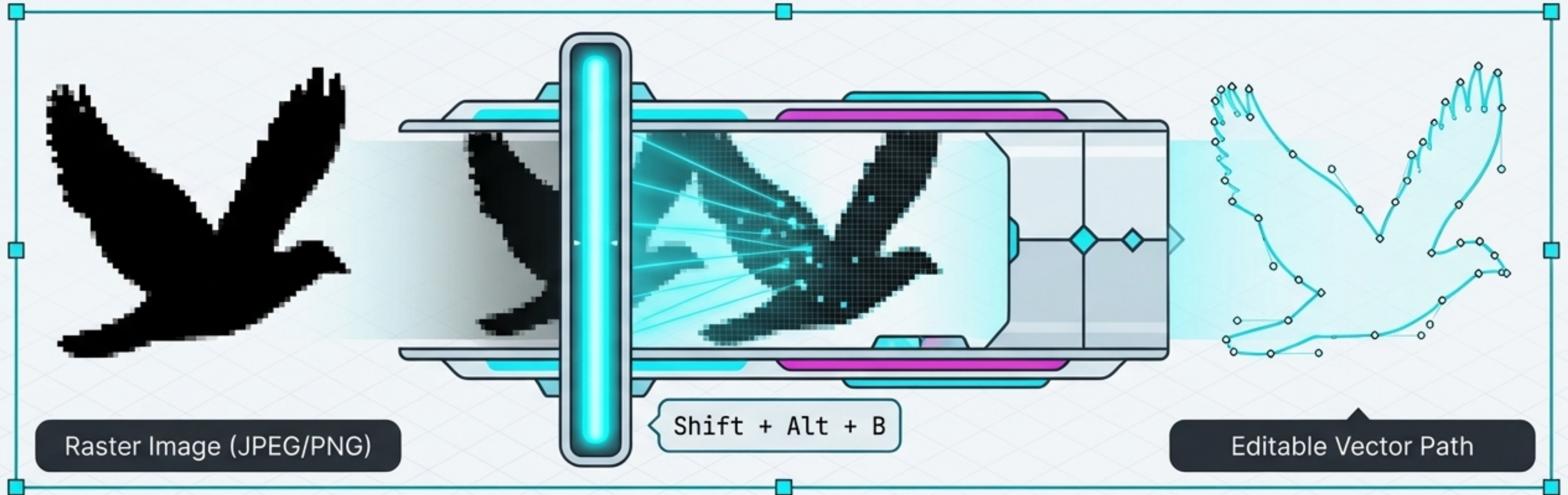
Hides everything outside the top shape using a **hard edge**. Perfect for strictly fitting photographs or textures into specific shapes like circles or letters.

Masking



Changes visibility dynamically using **tone and transparency**. Perfect for soft blends, fading images out, and creating cleaner, high-end editorial layouts.

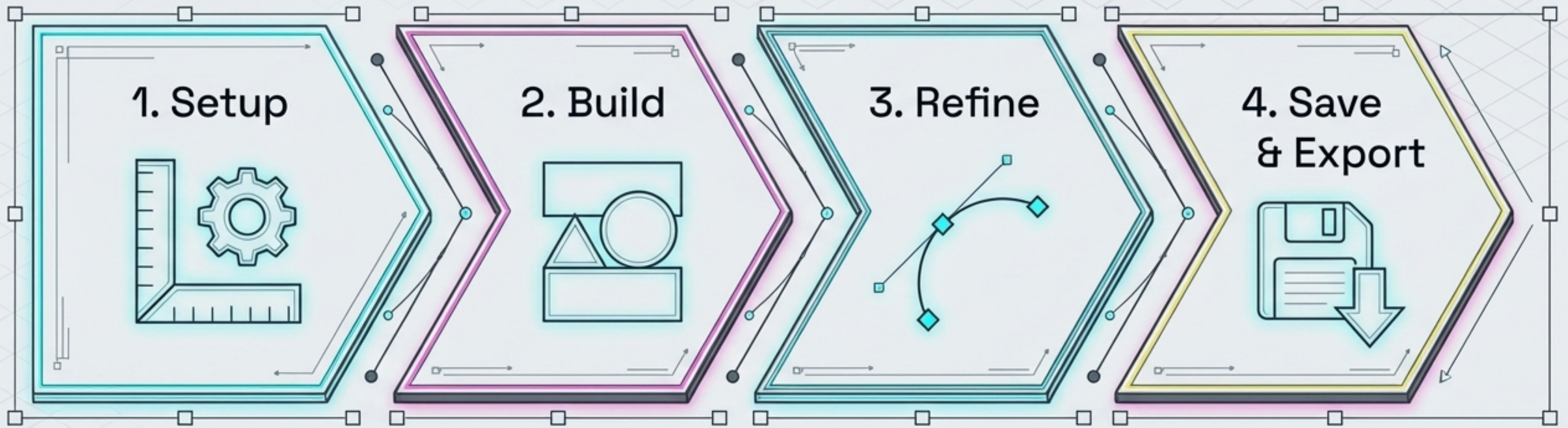
The Extraction Machine: Trace Bitmap



Inkscape can automatically trace a pixelated raster image and extract a clean, mathematical vector outline. Highly useful for digitising hand-drawn silhouettes or simple logos.

Warning: The extraction machine is not magic. The result is rarely perfect on complex photos and will require manual tidying using the Node tool.

The Designer's Workflow Lifecycle



Press **JetBrains Mono: Shift+Ctrl+D**. Set the exact page dimensions before you start drawing.

Use rigid Shapes, apply Fill/Stroke, and organise Layers. Rely on alignment tools and duplicate objects to keep the layout flawless.

Convert strictly necessary elements to Paths (**JetBrains Mono: Ctrl+Shift+C**). Edit nodes to construct unique, custom logo details.

Always save the master file as an SVG. Use "Save a Copy" to export a flat PDF for handing in to your teacher.

The KS4 Blueprint Checklist



Create a page of the correct size



Draw and colour basic shapes



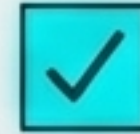
Add and format text



Duplicate objects (instead of redrawing)



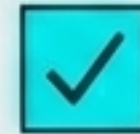
Align objects neatly



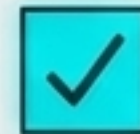
Change fill and stroke independently



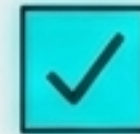
Convert a shape or text to a path



Edit nodes to customise a design



Trace a simple bitmap image



Export finished work while keeping the SVG safe

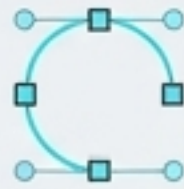
Keyboard Command Centre

Zone 1: File & Edit Basics

Ctrl + **C** / **Ctrl** + **V** : Copy / Paste

Ctrl + **D** : Duplicate in place

Shift + **Ctrl** + **V** : Paste style only



Zone 2: Layout & Organisation

Shift + **Ctrl** + **A** : Align and Distribute

Shift + **Ctrl** + **D** : Document Properties

Zone 3: Path & Shape Editing

Ctrl + **Shift** + **C** : Object to Path

Ctrl + **K** : Combine selected paths

Shift + **A** : Convert to Auto Nodes

Zone 4: Tracing & Drawing Tools

Shift + **Alt** + **B** : Trace Bitmap

C or **Ctrl** + **F6** : Calligraphy Tool

X or **Shift** + **F4** : 3D Box Tool